

Levantando Docker:

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % mkdir -p ~/docker/data/influxdb
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % docker compose up -d
[+] up 11/11
✓ Image influxdb:3-core Pulled 18.7s
✓ Network fixture2030-series-temporales_default Created 0.0s
✓ Container fixture2030-influxdb Started 0.7s
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % docker compose ps
NAME                IMAGE              COMMAND                                SERVICE    CREATED          STATUS
fixture2030-influxdb influxdb:3-core    "/usr/bin/entrypoint..."            influxdb    15 seconds ago   Restarting (255) Less than a second ago
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales %
```

Ver los logs:

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % docker compose logs --tail=50 influxdb
fixture2030-influxdb | 2026-10-07T14:22:03.964387Z WARN influxdb3: async triggers (run_asynchronous) currently default to unlimited concurrency; in a future release the default --async-trigger-concurrency-limit will change to 8. Set --async-trigger-concurrency-limit explicitly to keep the concurrency you want across upgrades.
fixture2030-influxdb | 2026-10-07T14:22:03.968108Z INFO influxdb3_lib::commands::serve: query concurrency limit computed from effective parallelism max_concurrent_queries=50
fixture2030-influxdb | 2026-10-07T14:22:03.970039Z INFO influxdb3_lib::commands::serve: server starting node_id=fixture2030-local git_hash=3ba97c65f1ee4e1f127a8266517d4d2083b7ea39 version=3.12.0 uuid=32bf7182-431b-4b0b-b381-1530bf44124e num_cpus=6 product_name="InfluxDB 3 Core"
fixture2030-influxdb | 2026-10-07T14:22:03.971460Z INFO influxdb3_clap_blocks::object_store: Object Store db_dir="/var/lib/influxdb3/data" object_store_type="Directory"
fixture2030-influxdb | 2026-10-07T14:22:03.977919Z INFO influxdb3_lib::commands::serve: initialising parquet cache
fixture2030-influxdb | 2026-10-07T14:22:03.978110Z INFO influxdb3_lib::commands::serve: Creating shared query executor num_threads=6
fixture2030-influxdb | 2026-10-07T14:22:04.017844Z INFO influxdb3_catalog::object_store::versions::v3: catalog not found, initializing a new one catalog_uuid=e1a4610f-79af-4a37-9bc1-47933ff52e24 prefix=fixture2030-local
fixture2030-influxdb | 2026-10-07T14:22:04.024698Z INFO influxdb3_catalog::catalog::versions::v3: catalog: create database name="_internal" options=CreateDatabaseOptions { retention_period: Some(604800s), schema_mode: Implicit }
fixture2030-influxdb | 2026-10-07T14:22:04.032926Z INFO influxdb3_catalog::catalog::versions::v3: catalog: created internal database
fixture2030-influxdb | 2026-10-07T14:22:04.033028Z INFO influxdb3_lib::commands::serve: catalog initialized catalog_uuid=e1a4610f-79af-4a37-9bc1-47933ff52e24
fixture2030-influxdb | 2026-10-07T14:22:04.033167Z INFO influxdb3_lib::commands::serve: Initializing table index cache node_id="fixture2030-local" max_entries=Some(100) concurrency_limit=20
fixture2030-influxdb | 2026-10-07T14:22:04.033369Z INFO influxdb3_write::table_index_cache: creating table indices from split snapshots
fixture2030-influxdb | 2026-10-07T14:22:04.034409Z INFO influxdb3_write::table_index_cache: loading all snapshot object metas
fixture2030-influxdb | 2026-10-07T14:22:04.035174Z INFO influxdb3_write::table_index_cache: Starting table index cache synchronization from object store node_prefix=fixture2030-local
fixture2030-influxdb | 2026-10-07T14:22:04.035450Z INFO influxdb3_write::table_index_cache: No table new table indices to be created. duration_ms=0 total_cached=0 indices_updated=0 cache_capacity=Some(100)
```

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % docker compose exec -l influxdb influxdb3 --help
InfluxDB 3 Core Server and Command Line Tools
```

```
# Example to run the InfluxDB 3 Core server:
influxdb3 serve --node-id my_node_name --object-store file --data-dir ~/.influxdb3_data
```

Usage: influxdb3 [OPTIONS] [COMMAND]

Common Commands:

serve	Run the InfluxDB 3 Core server
query, q	Perform a query against a running InfluxDB 3 Core server
write, w	Perform a set of writes to a running InfluxDB 3 Core server
update	Update resources on the InfluxDB 3 Core server

Resource Management:

create	Create a resource such as a database or auth token
show	List resources on the InfluxDB 3 Core server
delete	Delete a resource such as a database or table
enable	Enable a resource such as a trigger
disable	Disable a resource such as a trigger

System Management:

install	Install Python packages for the Processing Engine
test	Test that Processing Engine plugins work the way you expect

Common Options:

-h, --help	Print help information
-V, --version	Print version information

Advanced Help Options:

Use **--help-all** to see runtime configuration options

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales %
```

Iniciar entorno (no se incluye los tokens en la documentacion como fue requerido):

```

==== this complete error with server - server is not up - help me fix this complete error
[tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % bash scripts/inicializacion.sh
=== Esperando disponibilidad del servidor ===
=== Reutilizando token local existente ===
=== Version / ping ===
{"product_name":"InfluxDB 3 Core","version":"3.12.0","revision":"3ba97c65f1","process_id":"326ffeeb-2bb1-4b66-8922-df49af4c1dc3"}
=== Base de datos ===
Base fixture2030_metrics ya existe.
=== Tablas ===
Table "fixture2030_metrics"."estadisticas_partido" created successfully
Table "fixture2030_metrics"."usuarios_conectados" created successfully
=== SHOW TABLES ===
+-----+-----+-----+-----+
| table_catalog | table_schema | table_name | table_type |
+-----+-----+-----+-----+
| public        | iox           | estadisticas_partido | BASE TABLE |
| public        | iox           | usuarios_conectados  | BASE TABLE |
| public        | system        | distinct_caches      | BASE TABLE |
| public        | system        | influxdb_schema       | BASE TABLE |
| public        | system        | last_caches           | BASE TABLE |
| public        | system        | parquet_files         | BASE TABLE |
| public        | system        | processing_engine_logs | VIEW         |
| public        | system        | processing_engine_trigger_arguments | BASE TABLE |
| public        | system        | processing_engine_triggers | BASE TABLE |
| public        | system        | queries               | BASE TABLE |
| public        | information_schema | tables               | VIEW         |
| public        | information_schema | views                | VIEW         |
| public        | information_schema | columns              | VIEW         |
| public        | information_schema | df_settings          | VIEW         |
| public        | information_schema | schemata              | VIEW         |
| public        | information_schema | routines              | VIEW         |
| public        | information_schema | parameters            | VIEW         |
+-----+-----+-----+-----+

=== Retención ===
+-----+-----+
| database_name | retention_period |
+-----+-----+
| fixture2030_metrics | 90.0000d         |
+-----+-----+

=== Health ===
OK
Inicializacion OK. No publiques /Users/tomy/fixture2030-series-temporales/.secrets/influxdb_admin_token.
[tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % ]

```

Validar antes de carga de datos:

```

[tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % bash scripts/validacion.sh
=== SHOW TABLES ===
+-----+-----+-----+-----+
| table_catalog | table_schema | table_name | table_type |
+-----+-----+-----+-----+
| public        | iox           | estadisticas_partido | BASE TABLE |
| public        | iox           | usuarios_conectados  | BASE TABLE |
| public        | system        | distinct_caches      | BASE TABLE |
| public        | system        | influxdb_schema       | BASE TABLE |
| public        | system        | last_caches           | BASE TABLE |
| public        | system        | parquet_files         | BASE TABLE |
| public        | system        | processing_engine_logs | VIEW         |
| public        | system        | processing_engine_trigger_arguments | BASE TABLE |
| public        | system        | processing_engine_triggers | BASE TABLE |
| public        | system        | queries               | BASE TABLE |
| public        | information_schema | tables               | VIEW         |
| public        | information_schema | views                | VIEW         |
| public        | information_schema | columns              | VIEW         |
| public        | information_schema | df_settings          | VIEW         |
| public        | information_schema | schemata              | VIEW         |
| public        | information_schema | routines              | VIEW         |
| public        | information_schema | parameters            | VIEW         |
+-----+-----+-----+-----+

=== SHOW COLUMNS estadisticas_partido ===
+-----+-----+-----+-----+-----+-----+
| table_catalog | table_schema | table_name | column_name | data_type | is_nullable |
+-----+-----+-----+-----+-----+-----+
| public        | iox           | estadisticas_partido | equipo_id | Dictionary(Int32, Utf8) | YES |
| public        | iox           | estadisticas_partido | fase      | Dictionary(Int32, Utf8) | YES |
| public        | iox           | estadisticas_partido | fuente     | Dictionary(Int32, Utf8) | YES |
| public        | iox           | estadisticas_partido | partido_id | Dictionary(Int32, Utf8) | YES |
| public        | iox           | estadisticas_partido | sede       | Dictionary(Int32, Utf8) | YES |
| public        | iox           | estadisticas_partido | time       | Timestamp(ns) | NO |
+-----+-----+-----+-----+-----+-----+

=== SHOW COLUMNS usuarios_conectados ===
+-----+-----+-----+-----+-----+-----+
| table_catalog | table_schema | table_name | column_name | data_type | is_nullable |
+-----+-----+-----+-----+-----+-----+
| public        | iox           | usuarios_conectados | partido_id | Dictionary(Int32, Utf8) | YES |
| public        | iox           | usuarios_conectados | region     | Dictionary(Int32, Utf8) | YES |
| public        | iox           | usuarios_conectados | time       | Timestamp(ns) | NO |
+-----+-----+-----+-----+-----+-----+

=== Conteo estadísticas ===
+-----+
| puntos_estadisticas |
+-----+
| 0 |
+-----+

=== Conteo usuarios ===
+-----+
| puntos_usuarios |
+-----+
| 0 |
+-----+

=== Distribucion M001 por equipo ===
++
++

```

Cargar laboratorio:

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % python3 scripts/carga_lotes.py --matches 8 --batch-size 2000 --workers 2
=== Carga InfluxDB 3 Core ===
Partidos: 8
Batch size: 2000
Workers: 2
Puntos deportivos esperados: 640000
Puntos de usuarios esperados: 480
Puntos totales esperados: 640480
Precisión: second
Particionado: un worker procesa partidos completos para no mezclar la misma serie entre workers.
```

Validación de carga:

```
Partido 002 completado: 80000 puntos | 1/8 | 39.36s
Partido 001 completado: 80000 puntos | 2/8 | 39.37s
Partido 003 completado: 80000 puntos | 3/8 | 79.34s
Partido 004 completado: 80000 puntos | 4/8 | 79.36s
Partido 005 completado: 80000 puntos | 5/8 | 119.35s
Partido 006 completado: 80000 puntos | 6/8 | 119.35s
Partido 008 completado: 80000 puntos | 7/8 | 159.34s
Partido 007 completado: 80000 puntos | 8/8 | 159.34s
=== Resultado ===
Puntos deportivos cargados: 640000
Puntos usuarios cargados: 480
Puntos totales cargados: 640480
Tiempo observado (s): 160.327
Puntos/s observados: 3994.84
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % bash scripts/validacion.sh
=== SHOW TABLES ===
+-----+-----+-----+-----+
| table_catalog | table_schema | table_name | table_type |
+-----+-----+-----+-----+
| public        | iox          | estadisticas_partido | BASE TABLE |
| public        | iox          | usuarios_conectados  | BASE TABLE |
| public        | system       | distinct_caches      | BASE TABLE |
| public        | system       | influxdb_schema       | BASE TABLE |
| public        | system       | last_caches           | BASE TABLE |
| public        | system       | parquet_files         | BASE TABLE |
| public        | system       | processing_engine_logs | VIEW        |
| public        | system       | processing_engine_trigger_arguments | BASE TABLE |
| public        | system       | processing_engine_triggers | BASE TABLE |
| public        | system       | queries               | BASE TABLE |
| public        | information_schema | tables                | VIEW        |
| public        | information_schema | views                 | VIEW        |
| public        | information_schema | columns               | VIEW        |
| public        | information_schema | df_settings           | VIEW        |
| public        | information_schema | schemata              | VIEW        |
| public        | information_schema | routines              | VIEW        |
| public        | information_schema | parameters            | VIEW        |
+-----+-----+-----+-----+
=== SHOW COLUMNS estadisticas_partido ===
+-----+-----+-----+-----+-----+-----+
| table_catalog | table_schema | table_name | column_name | data_type | is_nullable |
+-----+-----+-----+-----+-----+-----+
| public        | iox          | estadisticas_partido | equipo_id | Dictionary(Int32, Utf8) | YES |
| public        | iox          | estadisticas_partido | fase      | Dictionary(Int32, Utf8) | YES |
| public        | iox          | estadisticas_partido | fuente     | Dictionary(Int32, Utf8) | YES |
| public        | iox          | estadisticas_partido | partido_id | Dictionary(Int32, Utf8) | YES |
| public        | iox          | estadisticas_partido | pases_completados | Int64      | YES |
| public        | iox          | estadisticas_partido | posesion_pct | Float64    | YES |
| public        | iox          | estadisticas_partido | recuperaciones | Int64      | YES |
| public        | iox          | estadisticas_partido | sede      | Dictionary(Int32, Utf8) | YES |
| public        | iox          | estadisticas_partido | time      | Timestamp(ns) | NO |
| public        | iox          | estadisticas_partido | tiros     | Int64      | YES |
| public        | iox          | estadisticas_partido | velocidad_kmh | Float64    | YES |
+-----+-----+-----+-----+-----+-----+
=== SHOW COLUMNS usuarios_conectados ===
```

```

=== Conteo estadísticas ===
+-----+
| puntos_estadísticas |
+-----+
| 640000 |
+-----+
=== Conteo usuarios ===
+-----+
| puntos_usuarios |
+-----+
| 480 |
+-----+
=== Distribucion M001 por equipo ===
+-----+-----+
| equipo_id | puntos |
+-----+-----+
| ARG      | 40000 |
| FRA      | 40000 |
+-----+-----+
=== Distribucion M001 por fuente ===
+-----+-----+
| fuente    | puntos |
+-----+-----+
| analitica | 10000 |
| app       | 10000 |
| arbitraje | 10000 |
| broadcast | 10000 |
| estadistica | 10000 |
| scouting  | 10000 |
| telemetria | 10000 |
| tracking  | 10000 |
+-----+-----+
=== Retencion ===
+-----+-----+
| database_name | retention_period |
+-----+-----+
| fixture2030_metrics | 90.0000d |
+-----+-----+
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales %

```

Consultas temporales:

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % bash scripts/consultas_temporales.sh
=== q01_ventana.sql ===
+-----+-----+-----+-----+-----+-----+-----+-----+
| time | partido_id | equipo_id | fuente | posesion_pct | pases_completados | tiros | recuperaciones | velocidad_kmh |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 2030-06-15T18:30:00 | M001 | ARG | analitica | 41.99 | 1 | 0 | 0 | 8.0 |
| 2030-06-15T18:30:00 | M001 | ARG | app | 43.39 | 3 | 0 | 0 | 8.4 |
| 2030-06-15T18:30:00 | M001 | ARG | arbitraje | 42.69 | 2 | 0 | 0 | 8.2 |
| 2030-06-15T18:30:00 | M001 | ARG | broadcast | 40.59 | 3 | 0 | 0 | 7.6 |
| 2030-06-15T18:30:00 | M001 | ARG | estadistica | 44.09 | 0 | 0 | 0 | 8.6 |
| 2030-06-15T18:30:00 | M001 | ARG | scouting | 39.19 | 1 | 0 | 1 | 7.2 |
| 2030-06-15T18:30:00 | M001 | ARG | telemetria | 41.29 | 0 | 0 | 0 | 7.8 |
| 2030-06-15T18:30:00 | M001 | ARG | tracking | 39.89 | 2 | 0 | 0 | 7.4 |
| 2030-06-15T18:30:01 | M001 | ARG | analitica | 44.58 | 0 | 0 | 0 | 15.7 |
| 2030-06-15T18:30:01 | M001 | ARG | app | 45.98 | 2 | 0 | 0 | 16.1 |
| 2030-06-15T18:30:01 | M001 | ARG | arbitraje | 45.28 | 1 | 0 | 0 | 15.9 |
| 2030-06-15T18:30:01 | M001 | ARG | broadcast | 43.18 | 2 | 0 | 0 | 15.3 |
| 2030-06-15T18:30:01 | M001 | ARG | estadistica | 46.68 | 3 | 0 | 1 | 16.3 |
| 2030-06-15T18:30:01 | M001 | ARG | scouting | 41.78 | 0 | 0 | 0 | 14.9 |
| 2030-06-15T18:30:01 | M001 | ARG | telemetria | 43.88 | 3 | 0 | 0 | 15.5 |
| 2030-06-15T18:30:01 | M001 | ARG | tracking | 42.48 | 1 | 0 | 0 | 15.1 |
| 2030-06-15T18:30:02 | M001 | ARG | analitica | 47.17 | 3 | 0 | 0 | 7.4 |
| 2030-06-15T18:30:02 | M001 | ARG | app | 48.57 | 1 | 0 | 0 | 7.8 |
| 2030-06-15T18:30:02 | M001 | ARG | arbitraje | 47.87 | 0 | 0 | 0 | 7.6 |
| 2030-06-15T18:30:02 | M001 | ARG | broadcast | 45.77 | 1 | 0 | 0 | 7.0 |
| 2030-06-15T18:30:02 | M001 | ARG | estadistica | 49.27 | 2 | 0 | 0 | 8.0 |
| 2030-06-15T18:30:02 | M001 | ARG | scouting | 44.37 | 3 | 0 | 0 | 6.6 |
| 2030-06-15T18:30:02 | M001 | ARG | telemetria | 46.47 | 2 | 0 | 0 | 7.2 |
| 2030-06-15T18:30:02 | M001 | ARG | tracking | 45.07 | 0 | 0 | 0 | 6.8 |
| 2030-06-15T18:30:03 | M001 | ARG | analitica | 49.76 | 2 | 0 | 1 | 15.1 |
| 2030-06-15T18:30:03 | M001 | ARG | app | 51.16 | 0 | 0 | 0 | 15.5 |
| 2030-06-15T18:30:03 | M001 | ARG | arbitraje | 50.46 | 3 | 0 | 0 | 15.3 |
| 2030-06-15T18:30:03 | M001 | ARG | broadcast | 48.36 | 0 | 0 | 0 | 14.7 |
| 2030-06-15T18:30:03 | M001 | ARG | estadistica | 51.86 | 1 | 0 | 0 | 15.7 |
| 2030-06-15T18:30:03 | M001 | ARG | scouting | 46.96 | 2 | 0 | 0 | 14.3 |
| 2030-06-15T18:30:03 | M001 | ARG | telemetria | 49.06 | 1 | 0 | 0 | 14.9 |
| 2030-06-15T18:30:03 | M001 | ARG | tracking | 47.66 | 3 | 0 | 0 | 14.5 |
| 2030-06-15T18:30:04 | M001 | ARG | analitica | 52.35 | 1 | 0 | 0 | 6.8 |
| 2030-06-15T18:30:04 | M001 | ARG | app | 53.75 | 3 | 0 | 0 | 7.2 |
| 2030-06-15T18:30:04 | M001 | ARG | arbitraje | 53.05 | 2 | 0 | 0 | 7.0 |
| 2030-06-15T18:30:04 | M001 | ARG | broadcast | 50.95 | 3 | 0 | 0 | 6.4 |
| 2030-06-15T18:30:04 | M001 | ARG | estadistica | 54.45 | 0 | 0 | 0 | 7.4 |
| 2030-06-15T18:30:04 | M001 | ARG | scouting | 49.55 | 1 | 0 | 0 | 6.0 |
| 2030-06-15T18:30:04 | M001 | ARG | telemetria | 51.65 | 0 | 0 | 0 | 6.6 |
| 2030-06-15T18:30:04 | M001 | ARG | tracking | 50.25 | 2 | 0 | 0 | 6.2 |
+-----+-----+-----+-----+-----+-----+-----+-----+

=== q02_comparacion_equipos.sql ===
+-----+-----+-----+-----+-----+-----+
| equipo_id | posesion_promedio | pases_totales | tiros_totales | recuperaciones_totales | velocidad_promedio |
+-----+-----+-----+-----+-----+-----+
| ARG | 50.330485416666666 | 7200 | 89 | 283 | 13.116666666666667 |
| FRA | 50.387516666666667 | 7200 | 91 | 283 | 13.15 |
+-----+-----+-----+-----+-----+-----+

=== q03_comparacion_fuentes.sql ===
+-----+-----+-----+-----+
| fuente | posesion_promedio | pases_totales | tiros_totales |
+-----+-----+-----+-----+
| analitica | 50.7172 | 1800 | 23 |
| app | 52.052299999999999 | 1800 | 23 |
| arbitraje | 51.392708333333336 | 1800 | 22 |
| broadcast | 49.333349999999998 | 1800 | 22 |
| estadistica | 52.695333333333334 | 1800 | 22 |
| scouting | 47.998383333333334 | 1800 | 22 |
| telemetria | 50.025 | 1800 | 23 |
| tracking | 48.657733333333326 | 1800 | 23 |
+-----+-----+-----+-----+

=== q04_pico_usuarios.sql ===
+-----+-----+
| region | pico_usuarios_activos |
+-----+-----+
| Patagonia | 17570 |
| NEA | 17067 |
| NOA | 16564 |
| Centro | 16061 |
| AMBA | 15558 |
+-----+-----+

tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales %
```

Agregaciones:

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % bash scripts/agregaciones.sh
=== ag01_equipo.sql ===
+-----+-----+-----+-----+-----+
| equipo_id | posesion_promedio | pases_totales | tiros_totales | velocidad_maxima |
+-----+-----+-----+-----+-----+
| ARG       | 50.33048541666666 | 7200           | 89            | 21.8              |
| FRA       | 50.38751666666667 | 7200           | 91            | 21.8              |
+-----+-----+-----+-----+-----+

=== ag02_por_minuto.sql ===
+-----+-----+-----+-----+-----+-----+
| minuto      | equipo_id | posesion_promedio | pases_totales | tiros_totales | velocidad_promedio |
+-----+-----+-----+-----+-----+-----+
| 2030-06-15T18:30:00 | ARG       | 50.0315          | 720           | 8             | 12.916666666666666 |
| 2030-06-15T18:30:00 | FRA       | 50.33497916666685 | 720           | 10            | 13.083333333333336 |
| 2030-06-15T18:31:00 | ARG       | 50.42189583333345 | 720           | 9             | 13.05               |
| 2030-06-15T18:31:00 | FRA       | 50.24847916666666 | 720           | 9             | 13.216666666666667 |
| 2030-06-15T18:32:00 | ARG       | 50.33502083333334 | 720           | 9             | 13.45               |
| 2030-06-15T18:32:00 | FRA       | 50.63839583333306 | 720           | 8             | 13.35               |
| 2030-06-15T18:33:00 | ARG       | 50.72522916666666 | 720           | 9             | 13.05               |
| 2030-06-15T18:33:00 | FRA       | 50.53597916666668 | 720           | 10            | 12.95               |
| 2030-06-15T18:34:00 | ARG       | 50.14072916666666 | 720           | 9             | 12.916666666666666 |
| 2030-06-15T18:34:00 | FRA       | 49.94024999999998 | 720           | 9             | 13.083333333333336 |
| 2030-06-15T18:35:00 | ARG       | 50.0268125        | 720           | 9             | 13.316666666666666 |
| 2030-06-15T18:35:00 | FRA       | 50.32993750000002 | 720           | 9             | 13.483333333333336 |
| 2030-06-15T18:36:00 | ARG       | 49.94031250000001 | 720           | 10            | 13.183333333333334 |
| 2030-06-15T18:36:00 | FRA       | 50.24347916666674 | 720           | 9             | 13.083333333333334 |
| 2030-06-15T18:37:00 | ARG       | 50.33014583333334 | 720           | 9             | 13.316666666666666 |
| 2030-06-15T18:37:00 | FRA       | 50.63364583333334 | 720           | 9             | 12.95               |
| 2030-06-15T18:38:00 | ARG       | 50.72049999999994 | 720           | 9             | 12.916666666666666 |
| 2030-06-15T18:38:00 | FRA       | 50.54706249999996 | 720           | 9             | 13.083333333333334 |
| 2030-06-15T18:39:00 | ARG       | 50.63270833333332 | 720           | 8             | 13.05               |
| 2030-06-15T18:39:00 | FRA       | 50.42295833333334 | 720           | 9             | 13.216666666666667 |
+-----+-----+-----+-----+-----+-----+

tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales %
```

Medición formal y evidencia:

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % python3 scripts/medir.py --matches 8 --batch-size 2000 --workers 2
```

```
tomy@MacBook-Neo-de-Tomy ~ % bash scripts/capturar_evidencia.sh
bash: scripts/capturar_evidencia.sh: No such file or directory
tomy@MacBook-Neo-de-Tomy ~ % cd fixture2030-series-temporales
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % bash scripts/capturar_evidencia.sh
Evidencia generada: /Users/tomy/fixture2030-series-temporales/docs/evidencia/ejecucion_20261007_173812.txt
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % ls -lh docs/evidencia
total 48
-rw-r--r-- 1 tomy staff 17K 7 oct. 14:38 ejecucion_20261007_173812.txt
-rw-r--r-- 1 tomy staff 544B 7 oct. 10:55 README.md
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales %
```

Consultas temporales:

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % bash scripts/consultas_temporales.sh
=== q01_ventana.sql ===
```

time	partido_id	equipo_id	fuelle	posesion_pct	pases_completados	tiros	recuperaciones	velocidad_kmh
2030-06-15T18:30:00	M001	ARG	analitica	41.99	1	0	0	8.0
2030-06-15T18:30:00	M001	ARG	app	43.39	3	0	0	8.4
2030-06-15T18:30:00	M001	ARG	arbitraje	42.69	2	0	0	8.2
2030-06-15T18:30:00	M001	ARG	broadcast	40.59	3	0	0	7.6
2030-06-15T18:30:00	M001	ARG	estadistica	44.09	0	0	0	8.6
2030-06-15T18:30:00	M001	ARG	scouting	39.19	1	0	1	7.2
2030-06-15T18:30:00	M001	ARG	telemetria	41.29	0	0	0	7.8
2030-06-15T18:30:00	M001	ARG	tracking	39.89	2	0	0	7.4
2030-06-15T18:30:01	M001	ARG	analitica	44.58	0	0	0	15.7
2030-06-15T18:30:01	M001	ARG	app	45.98	2	0	0	16.1
2030-06-15T18:30:01	M001	ARG	arbitraje	45.28	1	0	0	15.9
2030-06-15T18:30:01	M001	ARG	broadcast	43.18	2	0	0	15.3
2030-06-15T18:30:01	M001	ARG	estadistica	46.68	3	0	1	16.3
2030-06-15T18:30:01	M001	ARG	scouting	41.78	0	0	0	14.9
2030-06-15T18:30:01	M001	ARG	telemetria	43.88	3	0	0	15.5
2030-06-15T18:30:01	M001	ARG	tracking	42.48	1	0	0	15.1
2030-06-15T18:30:02	M001	ARG	analitica	47.17	3	0	0	7.4
2030-06-15T18:30:02	M001	ARG	app	48.57	1	0	0	7.8
2030-06-15T18:30:02	M001	ARG	arbitraje	47.87	0	0	0	7.6
2030-06-15T18:30:02	M001	ARG	broadcast	45.77	1	0	0	7.0
2030-06-15T18:30:02	M001	ARG	estadistica	49.27	2	0	0	8.0
2030-06-15T18:30:02	M001	ARG	scouting	44.37	3	0	0	6.6
2030-06-15T18:30:02	M001	ARG	telemetria	46.47	2	0	0	7.2
2030-06-15T18:30:02	M001	ARG	tracking	45.07	0	0	0	6.8
2030-06-15T18:30:03	M001	ARG	analitica	49.76	2	0	1	15.1
2030-06-15T18:30:03	M001	ARG	app	51.16	0	0	0	15.5
2030-06-15T18:30:03	M001	ARG	arbitraje	50.46	3	0	0	15.3
2030-06-15T18:30:03	M001	ARG	broadcast	48.36	0	0	0	14.7
2030-06-15T18:30:03	M001	ARG	estadistica	51.86	1	0	0	15.7
2030-06-15T18:30:03	M001	ARG	scouting	46.96	2	0	0	14.3
2030-06-15T18:30:03	M001	ARG	telemetria	49.06	1	0	0	14.9
2030-06-15T18:30:03	M001	ARG	tracking	47.66	3	0	0	14.5
2030-06-15T18:30:04	M001	ARG	analitica	52.35	1	0	0	6.8
2030-06-15T18:30:04	M001	ARG	app	53.75	3	0	0	7.2
2030-06-15T18:30:04	M001	ARG	arbitraje	53.05	2	0	0	7.0
2030-06-15T18:30:04	M001	ARG	broadcast	50.95	3	0	0	6.4
2030-06-15T18:30:04	M001	ARG	estadistica	54.45	0	0	0	7.4
2030-06-15T18:30:04	M001	ARG	scouting	49.55	1	0	0	6.0
2030-06-15T18:30:04	M001	ARG	telemetria	51.65	0	0	0	6.6
2030-06-15T18:30:04	M001	ARG	tracking	50.25	2	0	0	6.2

```
=== q02_comparacion_equipos.sql ===
```

equipo_id	posesion_promedio	pases_totales	tiros_totales	recuperaciones_totales	velocidad_promedio
ARG	50.33048541666667	7200	89	283	13.116666666666667
FRA	50.38751666666667	7200	91	283	13.15

```
=== q03_comparacion_fuentes.sql ===
```

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales %
```

Agregaciones temporales:

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % bash scripts/agregaciones.sh
=== ag01_equipo.sql ===
```

equipo_id	posesion_promedio	pases_totales	tiros_totales	velocidad_maxima
ARG	50.33048541666667	7200	89	21.8
FRA	50.38751666666667	7200	91	21.8

```
=== ag02_por_minuto.sql ===
```

minuto	equipo_id	posesion_promedio	pases_totales	tiros_totales	velocidad_promedio
2030-06-15T18:30:00	ARG	50.03149999999999	720	8	12.916666666666666
2030-06-15T18:30:00	FRA	50.334979166666685	720	10	13.083333333333336
2030-06-15T18:31:00	ARG	50.421895833333334	720	9	13.05
2030-06-15T18:31:00	FRA	50.248479166666666	720	9	13.216666666666667
2030-06-15T18:32:00	ARG	50.335020833333335	720	9	13.45
2030-06-15T18:32:00	FRA	50.6383958333333306	720	8	13.35
2030-06-15T18:33:00	ARG	50.725229166666664	720	9	13.05
2030-06-15T18:33:00	FRA	50.535979166666668	720	10	12.95
2030-06-15T18:34:00	ARG	50.1407291666666674	720	9	12.916666666666666
2030-06-15T18:34:00	FRA	49.940249999999998	720	9	13.083333333333336
2030-06-15T18:35:00	ARG	50.0268125	720	9	13.316666666666666
2030-06-15T18:35:00	FRA	50.329937500000002	720	9	13.483333333333336
2030-06-15T18:36:00	ARG	49.940312500000005	720	10	13.183333333333334
2030-06-15T18:36:00	FRA	50.2434791666666674	720	9	13.083333333333334
2030-06-15T18:37:00	ARG	50.330145833333336	720	9	13.316666666666666
2030-06-15T18:37:00	FRA	50.633645833333334	720	9	12.95
2030-06-15T18:38:00	ARG	50.720499999999997	720	9	12.916666666666666
2030-06-15T18:38:00	FRA	50.547062499999996	720	9	13.083333333333334
2030-06-15T18:39:00	ARG	50.632708333333326	720	8	13.05
2030-06-15T18:39:00	FRA	50.422958333333334	720	9	13.216666666666667

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales %
```

Medicion formal:

```
tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % python3 scripts/medir.py --matches 8 --batch-size 2000 --workers 2
=== MEDICIÓN REAL ===
Fecha UTC: 2026-10-07T17:50:21Z
Partidos: 8
Batch size: 2000
Workers: 2
Tiempo carga (s): 160.329
Tiempo consulta agregada (s): 0.471
--- Salida carga ---
=== Carga InfluxDB 3 Core ===
Partidos: 8
Batch size: 2000
Workers: 2
Puntos deportivos esperados: 640000
Puntos de usuarios esperados: 480
Puntos totales esperados: 640480
Precisión: second
Particionado: un worker procesa partidos completos para no mezclar la misma serie entre workers.
Partido 001 completado: 80000 puntos | 1/8 | 39.22s
Partido 002 completado: 80000 puntos | 2/8 | 39.23s
Partido 004 completado: 80000 puntos | 3/8 | 79.22s
Partido 003 completado: 80000 puntos | 4/8 | 79.23s
Partido 005 completado: 80000 puntos | 5/8 | 119.22s
Partido 006 completado: 80000 puntos | 6/8 | 119.23s
Partido 008 completado: 80000 puntos | 7/8 | 159.22s
Partido 007 completado: 80000 puntos | 8/8 | 159.22s
=== Resultado ===
Puntos deportivos cargados: 640000
Puntos usuarios cargados: 480
Puntos totales cargados: 640480
Tiempo observado (s): 160.204
Puntos/s observados: 3997.90

--- Salida consulta ---
equipo_id,promedio
ARG,50.33048541666667
FRA,50.38751666666667

tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales %
```

Evidencia automatica + persistencia:

```

[tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % bash scripts/capturar_evidencia.sh
Evidencia generada: /Users/tomy/fixture2030-series-temporales/docs/evidencia/ejecucion_20261007_175511.txt
[tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % ls -lh docs/evidencia
total 88
-rw-r--r-- 1 tomy staff 17K 7 oct. 14:38 ejecucion_20261007_173812.txt
-rw-r--r-- 1 tomy staff 17K 7 oct. 14:55 ejecucion_20261007_175511.txt
-rw-r--r-- 1 tomy staff 544B 7 oct. 10:55 README.md
[tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % docker compose restart
[+] restart 0/1
. Container fixture2030-influxdb Restarting
[tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED      STATUS      PORTS
fixture2030-influxdb influxdb:3-core      "/usr/bin/entrypoint..." influxdb    3 hours ago  Up 3 seconds 127.0.0.1:8181->8181/tcp
[tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % TOKEN="$(tr -d '\r\n' < .secrets/influxdb_admin_token)"
[tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % docker compose exec -T influxdb influxdb3 query \
--database fixture2030_metrics \
--token "$TOKEN" \
"SELECT COUNT(*) AS total FROM estadisticas_partido"
+-----+
| total |
+-----+
| 640000 |
+-----+
[tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % █

```

Validacion final:

```

=== Conteo estadisticas ===
+-----+
| puntos_estadisticas |
+-----+
| 640000 |
+-----+
=== Conteo usuarios ===
+-----+
| puntos_usuarios |
+-----+
| 960 |
+-----+
=== Distribucion M001 por equipo ===
+-----+
| equipo_id | puntos |
+-----+
| ARG      | 40000 |
| FRA      | 40000 |
+-----+
=== Distribucion M001 por fuente ===
+-----+
| fuente      | puntos |
+-----+
| analitica   | 10000 |
| app         | 10000 |
| arbitraje   | 10000 |
| broadcast   | 10000 |
| estadistica | 10000 |
| scouting    | 10000 |
| telemetria  | 10000 |
| tracking     | 10000 |
+-----+
=== Retencion ===
+-----+
| database_name      | retention_period |
+-----+
| fixture2030_metrics | 90.0000d         |
+-----+
[tomy@MacBook-Neo-de-Tomy fixture2030-series-temporales % █

```